

Rule 10 If the present population of a city is P and there is an increment or decrement of $R_1\%$, $R_2\%$ and $R_3\%$ in first, second and third year respectively, then

Population of city after 3 yr

$$= P \left(1 \pm \frac{R_1}{100} \right) \left(1 \pm \frac{R_2}{100} \right) \left(1 \pm \frac{R_3}{100} \right)$$

Note Use '+' sign for increases and '-' sign for decreases.

EXAMPLE 7 Population of a city in 2004 was 1000000. If in 2005 there is an increment of 15%, in 2006 there is a decrement of 35% and in 2007 there is

an increment of 45%, then find the population of city at the end of year 2007.

a. 1083875 b. 1083000 c. 1089000 d. 1135000

Sol a. Given that, $P = 1000000$, $R_1 = 15\%$, $R_2 = 35\%$ (decrease) and $R_3 = 45\%$
Population of city at the end of year 2007.

$$\begin{aligned} &= P \left(1 + \frac{R_1}{100} \right) \left(1 - \frac{R_2}{100} \right) \left(1 + \frac{R_3}{100} \right) \\ &= 1000000 \left(1 + \frac{15}{100} \right) \left(1 - \frac{35}{100} \right) \left(1 + \frac{45}{100} \right) \\ &= 1000000 \times \frac{115}{100} \times \frac{65}{100} \times \frac{145}{100} \\ &= 1083875 \end{aligned}$$

> PRACTICE EXERCISE

- If 90% of $A = 30\%$ of B and $B = x\%$ of A , the value of x is
(a) 700 (b) 600 (c) 300 (d) 1100
- When 40% of a number is added to 42, the result is the number itself. The number is
(a) 70 (b) 90 (c) 82 (d) 72
- If $x\%$ of y is $13x$, then the value of y is
(a) 880 (b) 1300 (c) 1200 (d) 700
- What is the number when 20% of number is 30% of 40?
(a) 90 (b) 80 (c) 60 (d) 50
- What is the number when increased by 20% becomes 300?
(a) 250 (b) 200 (c) 180 (d) 280
- If 50% of $(x - y) = 40\%$ of $(x + y)$, then what percent of x is y ?
(a) $10\frac{1}{9}\%$ (b) $11\frac{1}{9}\%$ (c) $13\frac{1}{9}\%$ (d) $21\frac{1}{9}\%$
- Two candidates fought an election one get 65% of the votes and won by 300 votes. The total number of votes polled in the election is
(a) 700 (b) 950 (c) 1000 (d) 900
- A man spends ₹ 3500 and saves $12\frac{1}{2}\%$ of his income. His monthly income (in ₹) is
(a) 4000 (b) 3800 (c) 4200 (d) 4500
- In an examination 52% of the candidates failed in English, 42% in Mathematics and 17% in both. The number of those who passed in both the subjects is
(a) 23% (b) 40% (c) 53% (d) 33%
- To pass an examination, a candidate needs 40% marks. All questions carry equal marks. A candidate just passed by getting 10 answers correct by attempting 15 of the total questions. How many questions are there in the examination?
(a) 25 (b) 30
(c) 40 (d) 45
- If salary of X is 20% more than salary of Y , then by how much percentage is salary of Y less than X ?
(a) 25 (b) 20
(c) $\frac{50}{3}$ (d) $\frac{65}{4}$
- Water contains $14\frac{2}{7}\%$ of hydrogen and the rest is oxygen. In 350 g of water, oxygen will be
(a) 300 g (b) 250 g
(c) 200 g (d) None of these
- The income of 'A' is 20% higher than that of 'B'. The income of 'B' is 25% less than of 'C'. What per cent less is A's income from 'C's income?
(a) 7% (b) 8%
(c) 10% (d) 12.5%
- 38 L of milk was poured into a tub and the tub was found to be 5% empty. To completely fill the tub, what amount of additional milk must be poured?
(a) 1 L (b) 2 L (c) 3 L (d) 4 L
- 10% of the inhabitants of a certain city left that city. Later on 10% of the remaining inhabitants of that city again left the city. What is the remaining percentage of population of that city?
(a) 80% (b) 80.4% (c) 80.4% (d) 81%

- 16.** If after 24% of wastage the net output of a coal-mine is 68400 quintals. Then, the total output of the coal-mine in quintals is
(a) 70000 (b) 90000 (c) 80000 (d) 89000
- 17.** A rise of 25% in the price of grapes compels a person to buy 1.5 kg of grapes less for ₹ 240. Then, the original price of grapes per kg is
(a) ₹ 40 (b) ₹ 32 (c) ₹ 30 (d) ₹ 28
- 18.** The price of an item is increased by 20% and then decreased by 20% the final price as compared to original price is
(a) 4% less (b) 4% more (c) 20% less (d) 20% more
- 19.** Sohan saves 14% of his salary while George saves 22%. If both gets the same salary and George saves ₹ 1540. Then, the salary of each of them is
(a) ₹ 9500 (b) ₹ 17000 (c) ₹ 7000 (d) ₹ 7500
- 20.** A's salary is half that of B. If A got a 50% rise in his salary and B got a 25% rise in his salary, then the percentage increase in combined salaries of both is
(a) 13% (b) $33\frac{1}{3}\%$ (c) 33% (d) 45%
- 21.** A man donated 4% of his salary to a charity and deposited 10% of the rest in the bank. If now he has ₹ 10800, then his income was
(a) ₹ 13500 (b) ₹ 14500
(c) ₹ 40000 (d) ₹ 12500
- 22.** Rohit saves 30% of his salary. When his expenses increased by 30%, he is able to save ₹ 1215 per month. His monthly salary is
(a) ₹ 13500 (b) ₹ 14500
(c) ₹ 30000 (d) ₹ 12500
- 23.** A man spends 75% of his income. If his income is increased by 20% and he increased his expenditure by 10%. His savings percentage is increased by
(a) 25% (b) 50% (c) 75% (d) 10%
- 24.** The price of wheat has increased by 60%. In order to restore to the original price, the new price must be reduced by
(a) 37.5% (b) 33% (c) 34% (d) 40%
- 25.** If the numerator of a fraction increased by 20% and its denominator be diminished by 10%. The value of the fraction is $\frac{16}{27}$, then the fraction is
(a) $\frac{4}{9}$ (b) $\frac{3}{2}$ (c) $\frac{3}{8}$ (d) $\frac{9}{4}$
- 26.** 1 L of water is evaporated from 6 L of a solution having 4% of sugar. The percentage of sugar in the remaining solution is
(a) 4% (b) 5% (c) $4\frac{4}{5}\%$ (d) $\frac{4}{5}\%$
- 27.** Two numbers are less than a third number by 30% and 40%, respectively. How much per cent is the second number less than the first?
(a) 35% (b) 36% (c) 14% (d) $14\frac{2}{7}\%$
- 28.** A sample of 5 L of glycerine is formed to be adulterated to the extent of 20%. Find how glycerine should be added to bring down percentage of impurity to 5%?
(a) 10 L (b) 25 L (c) 15 L (d) 20 L
- 29.** The daily wages of a worker increase by 20% but the number of hours worked by him also dropped by 20%. If originally he was getting ₹ 200 per week, his wages per week now is
(a) ₹ 160 (b) ₹ 192 (c) ₹ 210 (d) ₹ 198
- 30.** 140 L of a liquid contains 90% of acid and the rest water. How much water must be added to make the water 12.5% of the resulting mixture?
(a) 4 L (b) 10 L (c) 12 L (d) 3 L
- 31.** A person spends 30% of monthly salary on rent, 25% on food, 20% on children's education and 12% on electricity and the balance of ₹ 1040 on the remaining items. What is the monthly salary of the person?
(a) ₹ 8000 (b) ₹ 9000 (c) ₹ 9600 (d) ₹ 10600
- 32.** In an office, 40% of the staff is female and rest is male. 60% of the male and 40% of female voted for Ramesh. The percentage of votes Ramesh got was
(a) 24% (b) 42% (c) 50% (d) 52%
- 33.** In a class-X of 30 students, 24 passed in first class; in another class-Y of 35 students, 28 passed in first class. In which class was the percentage of students passed first class more?
(a) Class-X (b) Class-Y
(c) Both X and Y (d) None of these
- Directions** (Q. Nos. 34-36) *The population of the town is 126800. It increases by 15% in the 1st year due to increase crime in the city.*
- 34.** What is the population of the town at the end of 2nd year if the population decreases by 20% in the second year?
(a) 174984 (b) 135996 (c) 116656 (d) 145820
- 35.** What is the population of the town, if the population decrease by 15% at the end of 2nd year?
(a) 123749 (b) 123479
(c) 123947 (d) None of these
- 36.** What is the population of the town, if the population increases by 20% at the end of 2nd year?
(a) 194784 (b) 174984 (c) 179484 (d) 178494

> PREVIOUS YEARS' QUESTIONS

- 37.** A man losses 20% of his money. After spending 25% of the remainder, he has ₹ 480 left. What is the amount of money he originally had? **☑ 2012 I**
 (a) ₹ 600 (b) ₹ 720 (c) ₹ 800 (d) 840
- 38.** The price of an article is ₹ 25. After two successive cuts by the same percentage, the price becomes ₹ 20.25. If each time the cut was $x\%$, then **☑ 2012 II**
 (a) $x = 9$ (b) $x = 10$ (c) $x = 11$ (d) $x = 11.5$
- 39.** What is 5% of 50% of 500? **☑ 2012 II**
 (a) 12.5 (b) 25 (c) 1.25 (d) 6.25
- 40.** X, Y and Z had taken a dinner together. The cost of the meal of Z was 20% more than that of Y and the cost of the meal of X was $\frac{5}{6}$ as much as the cost of the meal of Z. If Y paid ₹ 100, then what was the total amount that all the three of them had paid? **☑ 2013 II**
 (a) ₹ 285 (b) ₹ 300
 (c) ₹ 355 (d) None of these
- 41.** A water pipe is cut into two pieces. The longer piece is 70% of the length of the pipe. By how much percentage is the longer piece longer than the shorter piece? **☑ 2013 II**
 (a) 140% (b) $\frac{400}{3}\%$
 (c) 40% (d) None of these
- 42.** If $m\%$ of $m + n\%$ of $n = 2\%$ of $(m \times n)$, then what percentage of m is n ? **☑ 2014 II**
 (a) 50% (b) 75%
 (c) 100% (d) Cannot be determined
- 43.** The price of a commodity increased by 5% from 2010 to 2011, 8% from 2011 to 2012 and 77% from 2012 to 2013. What is the average price increase (approximate) from 2010 to 2013? **☑ 2014 II**
 (a) 26% (b) 32% (c) 24% (d) 30%
- 44.** A person could save 10% of his income. But 2 yr later, when his income increased by 20%, he could save the same amount only as before. By how much percentage has his expenditure increased? **☑ 2015 I**
 (a) $22\frac{2}{9}\%$ (b) $23\frac{1}{3}\%$
 (c) $24\frac{2}{9}\%$ (d) $25\frac{2}{9}\%$
- 45.** 20% of a number when added to 20 becomes the number itself, then the number is **☑ 2015 II**
 (a) 20 (b) 25 (c) 50 (d) 80
- 46.** A's salary was increased by 40% and then decreased by 20%. On the whole A's salary is increased by **☑ 2015 II**
 (a) 60% (b) 40% (c) 20% (d) 12%
- 47.** In an election 10% of the voters on the voter list did not cast their vote and 60 voters cast their ballot papers blank. There were only two candidates. The winner was supported by 47% of total voters in the voter list and he got 308 votes more than his rival. The number of voters on the voter list is **☑ 2015 II**
 (a) 3600 (b) 6200 (c) 6028 (d) 6400
- 48.** The salary of a person is increased by 10% of his original salary. But he received the same amount even after increment. What is the percentage of his salary he did not receive? **☑ 2016 I**
 (a) 11% (b) 10% (c) $\frac{100}{11}\%$ (d) $\frac{90}{11}\%$
- 49.** The expenditure of a household for a certain month is ₹ 20000, out of which ₹ 8000 is spent on education, ₹ 5900 on food, ₹ 2800 on shopping and the rest on personal care. What percentage of expenditure is spent on personal care? **☑ 2016 I**
 (a) 12% (b) 16.5% (c) 18% (d) 21.8%

> ANSWERS

1	c	2	a	3	b	4	c	5	a	6	b	7	c	8	a	9	a	10	a
11	c	12	a	13	c	14	b	15	d	16	b	17	b	18	a	19	c	20	b
21	d	22	a	23	b	24	a	25	a	26	c	27	d	28	c	29	b	30	a
31	a	32	d	33	c	34	c	35	c	36	b	37	c	38	b	39	a	40	d
41	b	42	c	43	d	44	a	45	b	46	d	47	b	48	c	49	b		

HINTS AND SOLUTIONS

1. (c) Given, 90% of $A = 30\%$ of B

$$\frac{90A}{100} = \frac{30B}{100} \quad [\text{Rule 1}]$$

$$\Rightarrow \frac{A}{B} = \frac{3}{9} \Rightarrow B = 3A$$

$$\text{Now, } B = x\% \text{ of } A, 3A = \frac{x \cdot A}{100} \quad [\text{Rule 1}]$$

$$\therefore x = 300$$

Hence, the value of x is 300.

2. (a) Let the number be x ,
 then, $42 + 40\%$ of $x = x$

$$\Rightarrow x = 42 + \frac{40}{100}x \quad [\text{Rule 1}]$$

$$\Rightarrow \frac{3}{5}x = 42$$

$$\therefore x = \frac{42 \times 5}{3} = 70$$

Hence, the required number is 70.

3. (b) Given, $x\%$ of $y = 13x$

$$\Rightarrow \frac{x}{100}y = 13x \quad [\text{Rule 1}]$$

$$\therefore y = 13 \times 100 = 1300$$

4. (c) Let the number be x .
 By given condition, 20% of x

$$= 30\% \text{ of } 40$$

$$\Rightarrow \frac{20x}{100} = \frac{30 \times 40}{100} \Rightarrow 20x = 30 \times 40$$

$$\therefore x = 60$$

Hence, the required number is 60.

5. (a) Let the number be x .

$$\therefore x + \frac{20x}{100} = 300$$

$$\Rightarrow x = \frac{300 \times 100}{120} = 250$$

6. (b) Given, 50% of $(x - y) = 40\%$ of $(x + y)$

$$\frac{50}{100} \times (x - y) = \frac{40}{100} \times (x + y)$$

$$\Rightarrow 5x - 5y = 4x + 4y \quad [\text{Rule 1}]$$

$$\Rightarrow x = 9y \quad \dots (i)$$

 Let $r\%$ of $x = y \Rightarrow \frac{r}{100} \times x = y$

$$\Rightarrow \frac{r}{100} \times 9y = y \quad [\text{from Eq. (i)}]$$

$$\therefore r = \frac{100}{9} = 11\frac{1}{9}\%$$

7. (c) Let total number of votes polled be x .

$$\therefore 65\% \text{ of } x - 35\% \text{ of } x = 300$$

$$\frac{65}{100}x - \frac{35}{100}x = 300$$

$$\Rightarrow \frac{30}{100}x = 300 \Rightarrow x = \frac{300 \times 100}{30}$$

$$\therefore x = 1000 \text{ votes}$$

Hence, the total number of votes is 1000.

8. (a) Let monthly income be ₹ x .

$$\Rightarrow 87\frac{1}{2}\% \text{ of } x = ₹ 3500$$

$$\Rightarrow \frac{175}{2 \times 100}x = ₹ 3500$$

$$\therefore x = \frac{3500 \times 2 \times 100}{175} = ₹ 4000$$

9. (a) Number of candidates who passed in both the subject = $100 - (x + y - z)\%$

$$= 100 - (52 + 42 - 17)\%$$

$$= 23\%$$

10. (a) Let the total number of questions in examination be x .

By given condition, 40% of $x = 10$

$$\Rightarrow \frac{x \times 40}{100} = 10 \Rightarrow x = \frac{1000}{40} = 25$$

11. (c) Let the salary of y be ₹ x .

$$\therefore \text{Salary of } x = ₹ \frac{120x}{100} = ₹ \frac{6x}{5}$$

Difference of their salaries

$$= ₹ \left(\frac{6x}{5} - x \right) = ₹ \frac{x}{5}$$

$$\therefore \text{Required percentage} = \frac{x/5}{6x/5} \times 100\% = \frac{50}{3}\%$$

[Rule 3]

Shortcut Method

Here, $x = 20\%$

$$\therefore \text{Required percentage} = \left(\frac{x}{100 + x} \times 100 \right)\% \quad [\text{Rule 5}]$$

$$= \left(\frac{20}{100 + 20} \times 100 \right)\%$$

$$= \frac{2000}{120}\% = \frac{50}{3}\%$$

12. (a) Percentage of oxygen in water

$$= 100 - 14\frac{2}{7} = 100 - \frac{100}{7} = 85\frac{5}{7}\%$$

$$\text{Percentage of oxygen in 350 g of water} = 85\frac{5}{7}\% \text{ of } 350 = \frac{600}{7} \times \frac{350}{100} = 300 \text{ g}$$

13. (c) Let the income of C be ₹ x .

$$\therefore \text{Income of } B = x - x \times 25\% = x - \frac{25x}{100}$$

$$= x - \frac{x}{4} = \frac{3x}{4}$$

$$\text{and, the income of } A = \frac{3x}{4} + \frac{3x}{4} \times 20\%$$

$$= \frac{3x}{4} + \frac{3x}{4} \times \frac{1}{5} = \frac{9x}{5}$$

Now, difference between C 's salary and A 's salary

$$= x - \frac{9x}{10} = \frac{x}{10}$$

$\therefore$ The required percentage

$$= \frac{x}{10 \times x} \times 100\% = 10\%$$

14. (b) Amount of milk in tub = 38 L

$\therefore$ Tub found to be 5% empty

$\therefore$ Total quantity of milk in per cent = 95%

Now, completely fill the tub, total amount of additional milk

$$= \frac{38 \times 5}{95} = 2 \text{ L}$$

15. (d) Remaining percentage

$$= \left(1 - \frac{10}{100} \right) \left(1 - \frac{10}{100} \right)$$

$$= \frac{90}{100} \times \frac{90}{100} = 81\%$$

16. (b) Let total output be x quintal.

$\therefore$ Useful output = $100 - 24 = 76\%$

$$\Rightarrow \frac{76}{100}x = 68400$$

$$\Rightarrow x = \frac{68400 \times 100}{76}$$

$$\therefore x = 90000 \text{ quintal}$$

17. (b) Let original price of grapes be ₹ x .

$$\therefore \text{Increased price} = x + \frac{25}{100}x$$

$$= ₹ \frac{125}{100}x \quad \dots (i)$$

Increased price of 1.5 kg of grapes

$$= \frac{25}{100} \times 240 = ₹ 60$$

$$\therefore \text{Increased price of 1 kg} = \frac{60}{1.5}$$

$$= ₹ 40 \quad \dots (ii)$$

So, from Eqs. (i) and (ii), we have

$$\frac{125}{100}x = ₹ 40$$

$$\therefore x = \frac{40 \times 100}{125} = ₹ 32$$

18. (a) The price of item first increased by 20% and then decreased by 20%.

$$\therefore \text{Net effect} = \left(20 - 20 + \frac{20 \times (-20)}{100} \right)$$

$$= \frac{-400}{100} = -4\% \quad [\text{Rule 8}]$$

19. (c) Let salary of each of them be ₹ x .
George saves 22% of x and his saving amount is ₹ 1540.

$$\Rightarrow \frac{22}{100}x = 1540$$

$$\Rightarrow x = \frac{1540 \times 100}{22} = ₹ 7000$$

20. (b) Let A 's salary = x . Then,

$$B\text{'s salary} = 2x$$

$$\text{New salary of } A = 150\% \text{ of } x = \frac{3}{2}x$$

$$\text{Total salary of } B = 125\% \text{ of } 2x = \frac{5}{2}x$$

$$\text{Total combined salary} \\ = \left(\frac{3}{2}x + \frac{5}{2}x \right) = 4x$$

$$\therefore \text{Required increment in salary} \\ = \frac{4x - 3x}{3x} \times 100 = \frac{100}{3} = 33\frac{1}{3}\%$$

[Rule 4]

21. (d) Let total income be ₹ x .

$$\text{Income deposited} = 10\% \text{ of } \left[x - \frac{4}{100}x \right]$$

$$= \frac{10}{100} \left(x - \frac{4}{100}x \right) = \frac{96}{1000}x$$

$$\text{Remaining income} = ₹ 10800$$

$$\therefore \frac{4}{100}x + \frac{96x}{1000} + 10800 = x$$

$$\Rightarrow 10800 = x - \frac{136x}{1000}$$

$$\Rightarrow 10800 = \frac{864x}{1000} \Rightarrow x = \frac{10800 \times 1000}{864}$$

$$\therefore x = ₹ 12500$$

22. (a) Let the salary of Rohit be ₹ 100, then saving = ₹ 30

$$\text{Expenses} = ₹ 70$$

$$\text{New expenses} = (100 + 30)\% \text{ of } ₹ 70 \\ = ₹ 91$$

$$\text{New saving} = ₹ (100 - 91) = ₹ 9$$

$$\text{He saves ₹ 9, his salary} = ₹ 100$$

$$\text{If he saves ₹ 1215.}$$

$$\text{Then, his salary} = ₹ \left(\frac{100}{9} \times 1215 \right) \\ = ₹ 13500$$

23. (b) Let income of man be ₹ 100.

$$\text{Then, his expenditure} = ₹ 75$$

$$\text{and savings} = ₹ 25$$

$$\text{New income} = ₹ (100 + 20) = ₹ 120$$

$$\text{New expenditure} = ₹ (75 + 7.5) \\ = ₹ 82.50$$

$$\text{New saving} = ₹ (120 - 82.5) = ₹ 37.50$$

$$\text{Saving difference} = 37.50 - 25.0 = 12.5$$

$$\therefore \text{Percentage increase saving} \\ = \frac{12.5}{25} \times 100 = 50\%$$

24. (a) New price reduced by

$$= \left(\frac{x}{100 + x} \right) \times 100\% \quad [\text{Rule 7}]$$

$$= \left(\frac{60}{100 + 60} \right) \times 100\% = 37.5\%$$

25. (a) Let fraction be $\frac{x}{y}$.

$$\text{New fraction} = \frac{120\% \text{ of } x}{90\% \text{ of } y} = \frac{4x}{3y}$$

According to question,

[by given condition]

$$\frac{4x}{3y} = \frac{16}{27} \Rightarrow \frac{x}{y} = \frac{16}{27} \times \frac{3}{4} = \frac{4}{9}$$

26. (c) Amount of sugar in 6 L of solution

$$= \frac{4}{100} \times 6 = 0.24 \text{ L}$$

$$\text{After evaporation, sugar in 5 L} = 0.24 \text{ L}$$

$\therefore$ Percentage of sugar

$$= \left(\frac{0.24}{5} \times 100 \right) = 4\frac{4}{5}\%$$

27. (d) Let the third number be z .

$$\therefore \text{First number } x = \frac{(100 - 30)}{100} \times z = \frac{7z}{10}$$

Second number

$$y = \frac{(100 - 40)}{100} \times z = \frac{6z}{10}$$

Difference between first and second number

$$= (x - y) = \frac{7z}{10} - \frac{6z}{10} = \frac{z}{10}$$

Hence, the required percent [Rule 4]

$$= \frac{\frac{z}{10}}{\frac{7z}{10}} \times 100\% = \frac{100}{7}\% = 14\frac{2}{7}\%$$

28. (c) Glycerine in the given sample = 80% of 5 L

$$= \frac{80}{100} \times 5 = 4 \text{ L}$$

Let x L of glycerine be added, then

$$\frac{4 + x}{(5 + x)} \times 100 = 95$$

$$\Rightarrow 80 + 20x = 95 + 19x$$

$$\therefore x = 15 \text{ L}$$

29. (b) Increased wages of the worker = 200 + 20% of 200 = ₹ 240

Also, let he worked for x h.

$\therefore$ Reduced working hours

$$= x - \frac{20}{100}x = 0.80x$$

$$\text{Required wages} = \frac{240}{x} \times 0.80x = ₹ 192$$

30. (a) Water in the mixture = 10% of 140 L
 $= \frac{10}{100} \times 140 = 14 \text{ L}$

Let x L of water added in the mixture,
 then $\left(\frac{14 + x}{140 + x} \right) \times 100 = 12.5$

$$\Rightarrow 1400 + 100x = 1750 + 12.5x$$

$$\Rightarrow 87.5x = 350$$

$$\Rightarrow x = \frac{350}{87.5} = 4 \text{ L}$$

31. (a) Let the monthly salary of the person be ₹ x .

By given condition,

$$\frac{(100 - 30 - 25 - 20 - 12) \times x}{100} = 1040$$

$$\Rightarrow \frac{13x}{100} = ₹ 1040 \Rightarrow x = \frac{1040 \times 100}{13}$$

$$\Rightarrow x = ₹ 8000$$

Hence, the monthly salary of the person is ₹ 8000.

32. (d) Let total number of staff be 100.

$$\text{Female staff} = 40$$

$$\text{Male staff} = (100 - 40) = 60$$

Votes casted by females

$$= \frac{40}{100} \times 40 = 16$$

$$\text{Votes casted by males} = \frac{60}{100} \times 60 = 36$$

Votes casted by both males and females

$$= 16 + 36 = 52$$

$\therefore$ Percentage votes obtained = 52%.

33. (c) For class-X, let the student passed in first class = $a\%$.

Then, by condition given in question,

$$a\% \text{ of } 30 = 24 \Rightarrow \frac{a \times 30}{100} = 24$$

$$\therefore a = 80\%$$

Now, for class-Y let the student passed in first class = $b\%$.

According to the question,

$$b\% \text{ of } 35 = 28$$

$$\Rightarrow \frac{b}{100} \times 35 = 28$$

$$\therefore b = 80\%$$

Hence, both classes have equal percentage of students getting first class.

34. (c) Given, $R_1 = 15\%$ and $R_2 = 20\%$.

$\therefore$ Required population

$$= P \left(1 + \frac{R_1}{100} \right) \left(1 - \frac{R_2}{100} \right) \quad [\text{Rule 10}]$$

$$= 126800 \left(1 + \frac{15}{100} \right) \left(1 - \frac{20}{100} \right)$$